
Semiconductor Physics

Karl W. Böer • Udo W. Pohl

Semiconductor Physics

Second Edition

With 721 Figures and 100 Tables

 Springer

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Foreword

Materials donate their names to the great epochs of human civilizations. The Stone Age with its heavy, huge monuments, the Bronze Age with the beauty of carefully designed tools, finally the Iron Age with all the useful splendor of machinery – they all furnish the visible proof. Now, however, we live in the Age of Semiconductors. Strange and modest are their first two syllables: “semi” – not exactly convincing. And yet, these materials are essentially invisible, truly small, hidden, yet most important helpers in our activities every day.

Karl Boer was himself a major investigator and proponent for this new class of materials. He contributed strongly in a variety of topics for these novel members of condensed matter. He used to live in the western part of divided Berlin, but worked most efficiently in its eastern part, directed research, demonstrated applications, and actively assisted in spreading the knowledge of science and technology for all international communities. Later, in the United States, he became also one of the most successful pioneers of solar energy conversion – with semiconductors, of course.

His handbook documented Boer’s broad range of personal knowledge and experience. All the readers had thus a wonderfully rich collection of the multitude of phenomena, very well explained and carefully presented. It is now a stroke of good fortune for all those users that Udo Pohl, who also became a Berliner, took it upon himself to amplify and modernize the handbook, to critically screen the contents. No question: the bulwark of semiconductor knowledge lives on!

Hans J. Queisser

Preface to the Second Edition

The first edition of *Semiconductor Physics* was published in spring 2018, in time for Karl W. Böer's 92nd birthday and only a month before he died. The book has been well received by the scientific community. Semiconductor devices are ubiquitous in our daily life, and today the global semiconductor industry is a key driver for economic growth. Recent discoveries in semiconductor physics established new and intriguing fields, which will open new scopes for applications in the future. One such field refers to the electronic states in a topological insulator, which recently led to a new classification scheme for electronic phases in condensed matter and in principle enables dissipationless surface conductance. Another field is quantum optics with the realization of photonic qubits based on excitonic states; rapid progress presently opens prospects for realizing quantum communication and quantum computing. Much of these developments are intimately related to the basic principles of semiconductors physics; some of the new topics are addressed in the second edition, keeping the style and the conciseness of the presentation. In addition, the initial text is thoroughly amended and updated.

Berlin
January 2023

Udo W. Pohl

Preface to the First Edition

The roots of this book can be traced back to the 1950s when one of us (KWB) started giving lectures on a young and intriguing emerging field – the physics of semiconductors. Semiconductor technology was in its infancy at that time, and the other of us (UWP) was just born. The accumulated and steadily updated lecture notes provided a broad background on semiconductor physics, and eventually – in 1990 – led to the first edition of the *Survey of Semiconductor Physics*. The textbook became a helpful source for generations of students of physics and electrical engineering. This applies as well for the younger of us two, who appreciated the concise treatment of the new book for preparing his lectures held on semiconductor physics and epitaxy.

Now, a quarter of a century later, semiconductor physics is still young and intriguing. New fields and materials joined the established topics, such as structures employing quantum effects and designed light-matter interaction, or nitride and organic semiconductors. Still their study bases upon the principles learned in the early lectures.

Our common roots in Berlin and our pleasure on teaching brought us together – by the amicable encouragement of Claus Ascheron from Springer. In this book we like to keep the conciseness of the presentation also given in the *Survey* and organize new topics in the same style. We had to select the material for keeping the volume to a manageable size; furthermore, like any book covering a large spectrum of subjects, the treatment lacks the depth of more focused texts. We apologize to the reader who may not find his own specialized subject represented in sufficient detail. This applies also to the references, which, in a steadily developing field, may not remain best choices; they essentially represent examples, helping to familiarize with the field and find more recent literature.

In the course of writing Dieter Palme steadily supported us by furnishing hundreds of illustrations for this book; we are much obliged for his precise work and critical comments for graphical representations. We appreciate being granted permission by publishers and authors for using illustrations from other published source, which were adapted and redrawn for a consistent presentation. Last but not least, we gratefully acknowledge Hans-Joachim Queisser for writing his concise foreword.

Naples
Berlin
January 2018

Karl W. Böer
Udo W. Pohl

Contents

Volume 1

Part I Growth, Bonding and Structure	1
1 Properties and Growth of Semiconductors	3
2 Crystal Bonding	35
3 The Structure of Semiconductors	59
Part II Phonons	111
4 Elasticity and Phonons	113
5 Phonon-Induced Thermal Properties	157
Part III Energy Bands	191
6 The Origin of Band Structure	193
7 Quantum Mechanics of Electrons in Crystals	219
8 Bands and Bandgaps in Solids	257
9 Magnetic Semiconductors	319
Part IV Photons	367
10 Interaction of Light with Solids	369
11 Photon–Phonon Interaction	429
12 Photon–Free-Electron Interaction	467
13 Band-to-Band Transitions	499
14 Excitons	529

Volume 2

Part V Defects	593
15 Crystal Defects	595
16 Crystal Interfaces	649
17 Optical Properties of Defects	703
18 Shallow-Level Centers	755
19 Deep-Level Centers	811
20 Defects in Amorphous and Organic Semiconductors	861
Part VI Transport	897
21 Equilibrium Statistics of Carriers	899
22 Carrier-Transport Equations	933
23 Carrier Scattering at Low Electric Fields	985
24 Carrier Scattering at High Electric Fields	1023
25 Carriers in Magnetic Fields and Temperature Gradients	1055
26 Superconductivity	1087
27 Carrier Transport in Low-Dimensional Semiconductors	1113
28 Carrier Transport Induced and Controlled by Defects	1165
Part VII Generation - Recombination	1203
29 Carrier Generation	1205
30 Carrier Recombination and Noise	1241
31 Photoconductivity	1299
32 Dynamic Processes	1325
Appendix: High-Field Domains	1381
Index	1403

About the Authors



Dr. mult. Karl Wolfgang Böer was distinguished professor of physics and solar energy, and fellow of AAAS, APS, ASES, and IEEE. Over five decades, he gave classes in solid state and semiconductor physics at the Humboldt University and the University of Delaware and gave keynote addresses at numerous international conferences. In his research, he was the first in measuring the Franz-Keldysh effect, discovered high-field domains, the condensation of excitons, and numerous properties of cadmium sulfide. He published 380 papers, 14 monographs, was editor of 23 international conference proceedings and obtained 18 patents. He created *Physica Status Solidi* in 1960 and the Series *Advances in Solar Energy* in 1978. He obtained highest honors from the University of Delaware, the American Solar Energy Society, and the International Solar Energy Society. The University of Delaware issued the prestigious solar energy award in his name that is given biannually to one of the world's leaders in solar energy research and industry. Karl Wolfgang Böer passed away aged 92 on April 18, 2018, shortly after the publication of the first edition of *Semiconductor Physics*.



Dr. rer. nat. Udo W. Pohl, professor of experimental physics, studied physics in Aachen and Berlin, Germany, and received his Ph.D. degree in 1988 from the Technical University of Berlin, where he was principal investigator in the Institute of Solid State Physics and retired end 2020. He is a member of the German Physical Society, participated in international conference committees, and has given keynote addresses at international conferences. Since the early 1990s, he has given lectures on semiconductor physics and epitaxy. In 2009,

he was appointed Adjunct Professor of Physics at the Technical University of Berlin. He is series editor of Materials Sciences, has authored over 200 journal articles and conference papers, the monograph *Epitaxy of Semiconductors*, 13 book contributions, and patents. His research interests include physics and epitaxy of semiconductor nanostructures and devices.



Karl W. Böer and Udo W. Pohl celebrating Karl's 91st birthday in March 2017 during the finalization of the first edition of *Semiconductor Physics*.